

simplicio-cli - 4-Quadrant Benchmark

Date: 2026-05-26

Models: google/gemma-3-4b-it, meta-llama/llama-3.2-3b-instruct, qwen/qwen-2.5-7b-instruct,
anthropic/claude-3.5-haiku

Cases: 10 | max_iters: 5 | base: <https://openrouter.ai/api/v1>

Why this matters

The original bench answered one question: does wrapping a goal in simplicio's 6-layer contract help a single LLM call? Answer: yes. But real-world coding agents add a retry loop on top. So the open question is: when you add a loop, does the prompt still matter, or does the loop alone close the gap? This bench answers it with a 2x2 matrix isolating prompt structure x execution model.

The matrix

Cell	Prompt	Execution
Q1	raw goal	1-shot (baseline)
Q2	simplicio 6-layer	1-shot (current bench)
Q3	raw goal	loop with feedback (<= 5 iters)
Q4	simplicio 6-layer	loop with feedback (composition)

Headline

Quadrant	Pass	Avg iters	Tokens / pass	ms / pass
Q1 no agent, no simplicio	0%	1.00	29,522	1,016,220
Q2 no agent, with simplicio	55%	1.00	991	23,551
Q3 with agent, no simplicio	47%	3.92	4,978	107,562
Q4 with agent, with simplicio	70%	2.60	1,563	30,584

Contribution decomposition

Delta	Formula	Value (pts)
Prompt effect, no loop	Q2 - Q1	+55
Loop effect, no simplicio	Q3 - Q1	+47
Prompt effect inside loop	Q4 - Q3	+23
Loop effect with simplicio	Q4 - Q2	+15
Composition over best single axis	Q4 - max(Q2, Q3)	+15
Synergy vs linear stacking	Q4 - linear	-32

Hypothesis verdicts (threshold $|\text{delta}| \geq 5$ pts)

- 1) Loop alone closes the gap: $Q4 - Q3 = +23$ pts -> REJECTED.
- 2) Simplicio alone is enough: $Q4 - Q2 = +15$ pts -> REJECTED.
- 3) Linear stacking (no synergy): $Q4 - \text{linear} = -32$ pts -> REJECTED.

Cost

Quadrant	Total tokens	Wall-clock (s)	Tokens / pass	ms / pass
Q1	29,522	1016.2	29,522	1,016,220
Q2	21,817	518.1	991	23,551
Q3	94,599	2043.7	4,978	107,562
Q4	43,791	856.4	1,563	30,584

Structural quality (rate across all runs)

Quadrant	DIFF block	TEST block	target mentioned
Q1	0%	65%	2%
Q2	75%	65%	77%
Q3	47%	70%	57%
Q4	75%	72%	77%

Per-model x quadrant

Model	Q1	Q2	Q3	Q4
google/gemma-3-4b-it	0/10	7/10	4/10	8/10
meta-llama/llama-3.2-3b-instruct	0/10	5/10	4/10	6/10
qwen/qwen-2.5-7b-instruct	0/10	6/10	8/10	10/10
anthropic/claude-3.5-haiku	0/10	4/10	3/10	4/10

Per-case x quadrant

#	Stack	Goal	Q1	Q2	Q3	Q4
1	angular	Hide the Delete button when the current user is not an	0/4	3/4	2/4	3/4
2	angular	Disable the email field unless the profile role is edit	0/4	3/4	2/4	4/4
3	angular	Only show the audit log link for users with role 'audit	0/4	3/4	2/4	3/4
4	angular	Show 'Approve' button only when the order status is 'pe	0/4	4/4	3/4	4/4
5	react	Render the export menu item only for users in the 'anal	0/4	1/4	3/4	2/4
6	react	Disable the 'Save Draft' button while the form is inval	0/4	3/4	1/4	3/4
7	react	Show a 'No results' empty state when the search returns	0/4	2/4	2/4	3/4
8	dotnet	Require the 'CanApprove' policy on the Approve endpoint	0/4	1/4	1/4	3/4
9	dotnet	Restrict the GET /reports endpoint so only users in the	0/4	2/4	2/4	2/4
10	angular	Show a warning banner if the user has unsaved changes	0/4	0/4	1/4	1/4

Reproduce

`pip install -e "[bench]"`

`export OPENROUTER_API_KEY=...`

`BENCH_MODELS="google/gemma-3-4b-it,meta-llama/llama-3.2-3b-instruct,qwen/qwen-2.5-7b-instruct,anthropic/claude-3.5-haiku" BENCH_MAX_ITERS=5 python3 bench/run_4quadrant.py`

Methodology: docs/benchmark-4quadrant.md

Raw outputs: .simplicio/bench_4q/<model>/case_NN/q*_iter*.txt